

KF SURYA

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Work Experience

- Gen AI Developer** Sep 2025 – Apr 2026
Connected Value Health (ValueHealthAI) Coimbatore, Tamil Nadu, India
 - Built a **Batch API pipeline** with Mistral OCR for PDF processing under the **Patient Nexus EBV** system, cutting per-page cost by 25% (\$1.00 → \$0.75 per 10 pages).
 - Trained a **Gradient Boosting model** to predict drug coverage status from drug type, manufacturer, and hidden policy requirements in insurance benefit documents.
- AI Intern** Jul 2024 – Aug 2024
Ei Systems Ghaziabad, Delhi NCR, India
 - Built **MediSaga RAG Chatbot** using RAPTOR indexing, Milvus, and quantized Llama2 for CPU-efficient medical QA.
- ML Intern** Dec 2023 – Jan 2024
Yanne Technologies Coimbatore, Tamil Nadu, India
 - Developed a **LangChain + GPT-3.5 Medical Assistant** for information retrieval and an **OpenCV Checkbox Detector** for Vizhi Eye Care NGO.

Education

- KPR Institute of Engineering and Technology** 2022 – 2026
B.Tech in Artificial Intelligence and Data Science Coimbatore, India
 - CGPA: 8.0/10

Technical Projects

- TruthLens — AI Fake News & Misinformation Detector** 2025
PyTorch, FastMCP, FastAPI, BERT, BiLSTM, React, TailwindCSS, MongoDB Atlas GitHub | Live
 - Built a **multimodal misinformation detection system** classifying text and image (OCR) content as Real / Fake / Uncertain with confidence scores and LLM-generated rationales.
 - Fine-tuned a **BERT + BiLSTM classifier** orchestrated via an **MCP server**; integrated live web search to cross-check claims and archived results in **MongoDB Atlas**.
- Multimodal RAG Chatbot** GitHub
FastAPI, Streamlit, ChromaDB, Vosk, Gemini 2.5 Flash, Docker
 - Built a **multimodal RAG system** supporting PDF, audio, and video inputs with **Dockerized deployment**.
 - Integrated **Gemini 2.5 Flash** and **Vosk speech-to-text** for context-aware AI responses at scale.
- Text-to-SQL Assistant** GitHub
Python, Chainlit, Gemini 2.5 Flash, SQLite
 - Developed an **AI-powered Text-to-SQL tool** converting natural language to SQL with schema detection, validation, and real-time result display.
- Company Expense Tracker** GitHub
Python 3.11, FastMCP, Gemini 2.5 Flash, SQLite
 - Built a **MCP server** enabling AI-driven expense and employee management with Gemini-based analytics.
- MediSaga — RAG Chatbot Application** GitHub
RAPTOR Indexing, Milvus, Llama2, Streamlit
 - Created a **medical QA chatbot** using RAPTOR indexing with Milvus and Llama2 for accurate retrieval.

Technical Skills

- Programming Languages:** Python, JavaScript, Java, C, SQL, Bash
- AI & Machine Learning:** RAG, LLM Integration, Transformers, Fine-Tuning, Model Quantization, Gradient Boosting, Classification
- NLP & Multimodal AI:** LangChain, Llama2/3, Gemini, BERT, BiLSTM, Sentence-Transformers, ChromaDB, FAISS, Vosk
- Frameworks & Tools:** FastAPI, FastMCP, Streamlit, Chainlit, PyTorch, TensorFlow, Hugging Face Transformers, Docker, Conda
- Databases:** MongoDB Atlas, PostgreSQL, SQLite, Milvus, Pinecone, ChromaDB
- Data Engineering:** Pandas, NumPy, Matplotlib, Batch Processing Pipelines, OCR (Mistral)
- Computer Vision:** OpenCV, YOLOv5/v8, Object Detection, Image Segmentation
- DevOps & Version Control:** Git, GitHub, Docker Compose, Environment Automation

Certifications & Awards

- Best Paper Award (AMSAS-2022)** — Certified by DRDO & Dept. of Chemistry, KPRIET
- Machine Learning with Python — IBM (Honors)
- Exploratory Data Analysis — Coursera
- Python for Data Analysis — Coursera
- Git & GitHub — 365 Data Science